



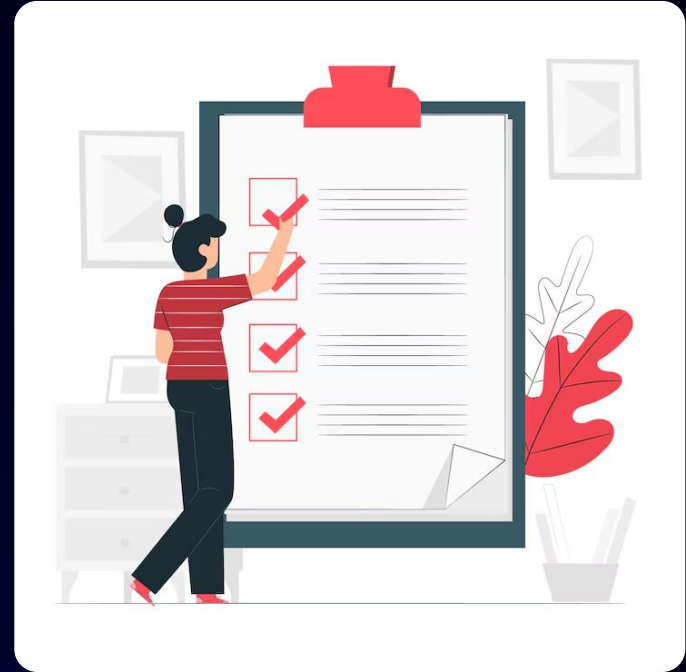
CSS Layout & Interaction – Box Model, Display, and Pseudo-Classes

Learning Objectives

- Understand the CSS Box Model (margin, border, padding, and content)
- Apply spacing effectively to improve layout and readability
- Understand the difference between visibility and display: none
- Gain a basic understanding of CSS specificity
- Add interactivity using pseudo-classes (:hover, :active, :focus)
- Improve an existing project using layout and styling techniques

Prerequisites

- Basic understanding of programming
- Basic understanding of HTML and CSS

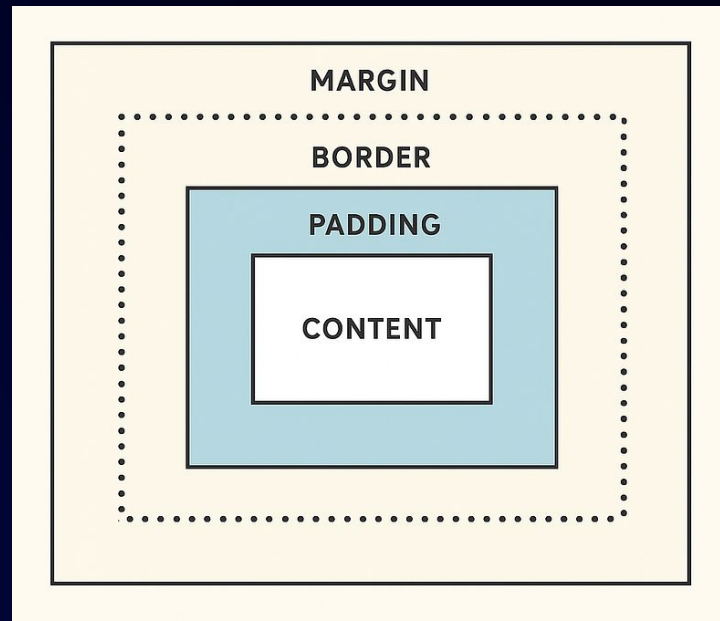




Box Model (Core Concept)

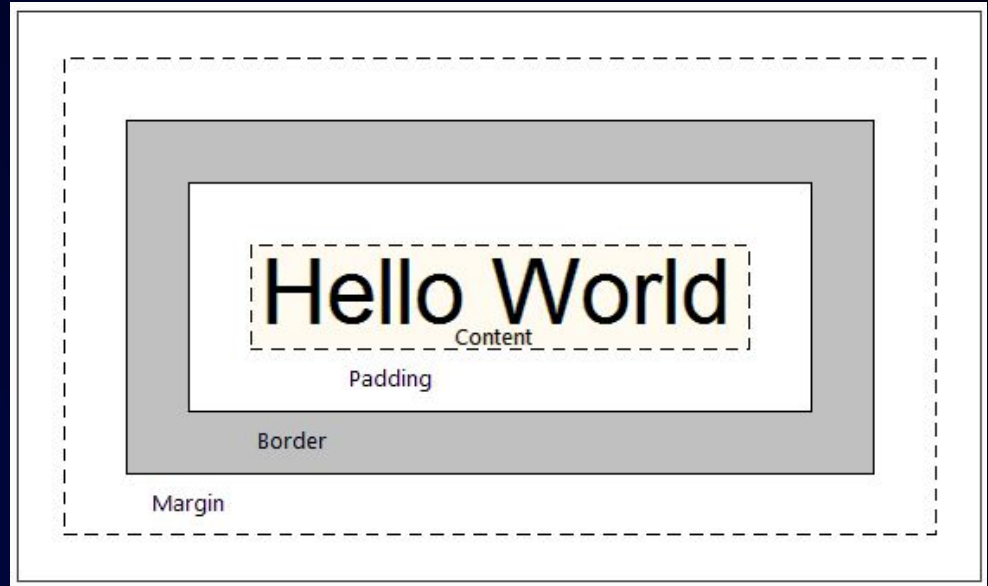
What is the Box Model?

- Every HTML element is displayed as a rectangular box.

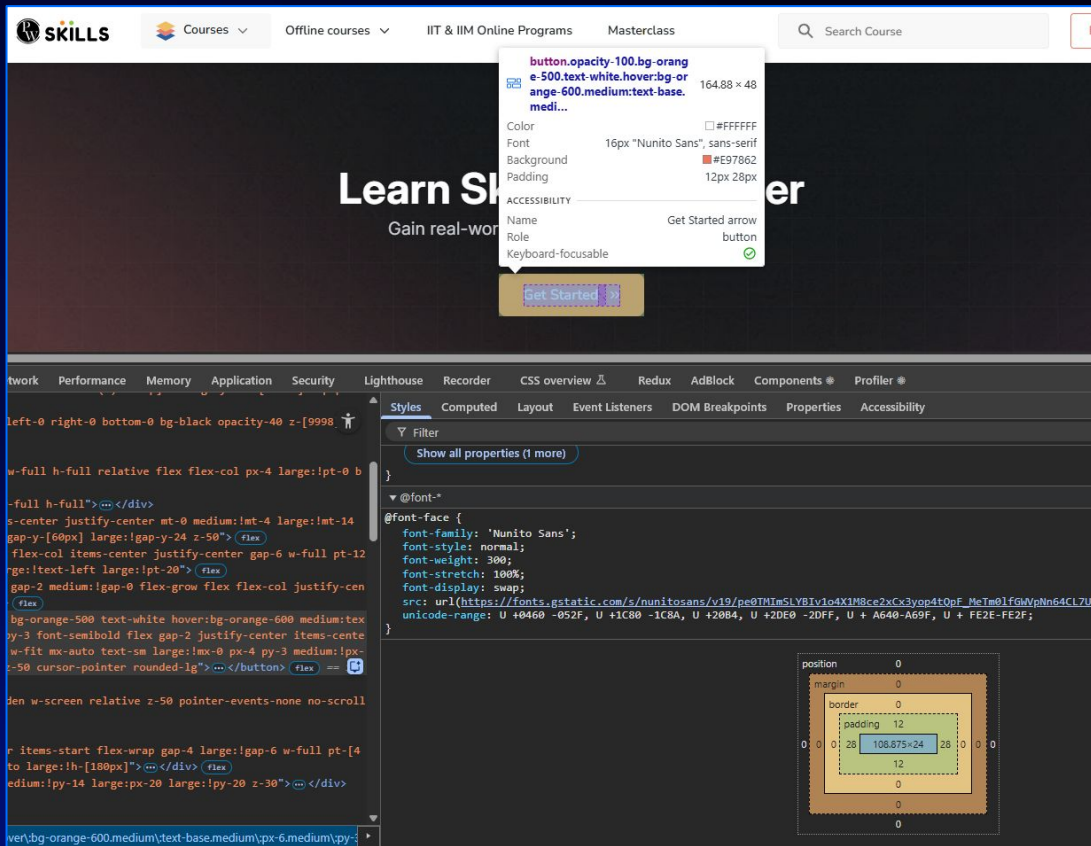


Components of the Box Model:

- **Content** – Actual text or image inside the element
- **Padding** – Space inside the box (around content)
- **Border** – Outline surrounding padding and content
- **Margin** – Space outside the box (between elements)



Visual Understanding: DevTools



The screenshot displays the DevTools interface with the 'Styles' panel open. The 'Computed' tab is selected, showing the following CSS rules for a 'button' element:

- `button.opacity-100.bg-orange`
- `e-500.text-white.hover:bg-orange-600.medium:text-base.medium:`

The 'Color' property is set to `#FFFFFF`. The 'Font' property is set to `16px "Nunito Sans", sans-serif`. The 'Background' property is set to `#E97862`. The 'Padding' property is set to `12px 28px`.

The 'Accessibility' panel shows the following information:

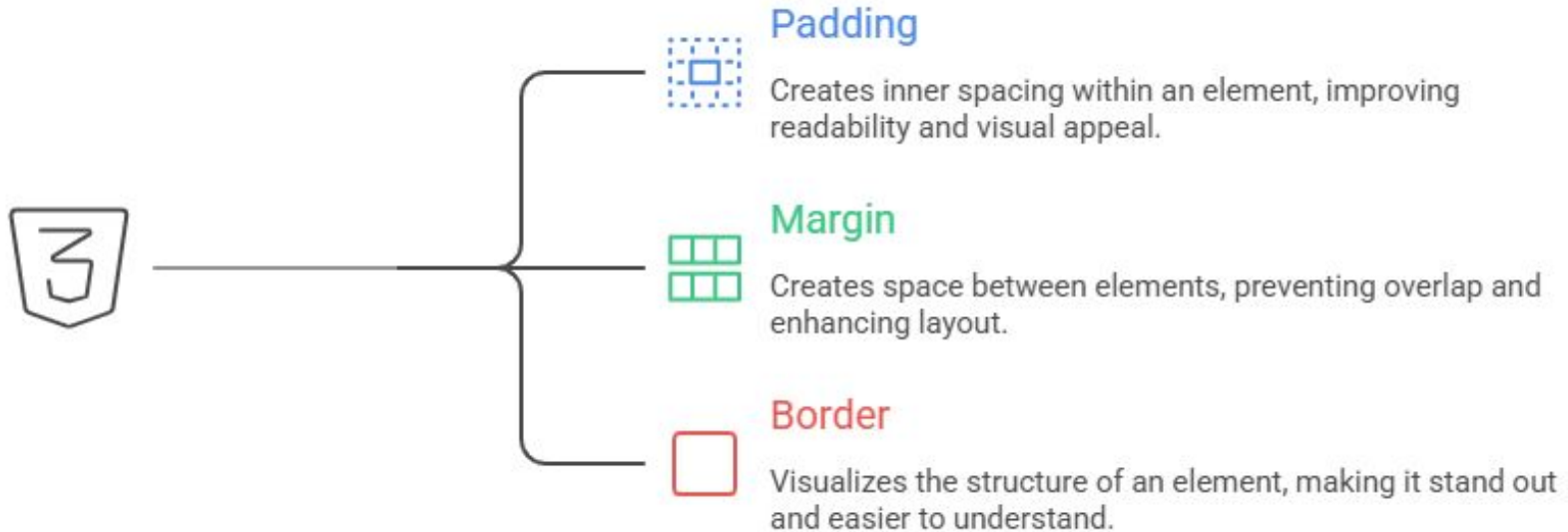
- Name: Get Started arrow
- Role: button
- Keyboard-focusable: ☒

The 'Visual' panel shows a diagram of the box model for the 'button' element. The diagram includes the following dimensions:

- width: 164.88 x 48
- height: 164.88 x 48
- padding: 12px 28px
- margin: 0px 0px 0px 0px

The 'Visual' panel also shows the 'position' property set to `relative` and the 'z-index' property set to `0`.

Applying the Box Model:



Hands-On Demo | Box Model (Core Concept)

Q. Which tool helps you visually inspect the box model?

A

VS Code

B

Browser DevTools

Q. Which tool helps you visually inspect the box model?

A

VS Code

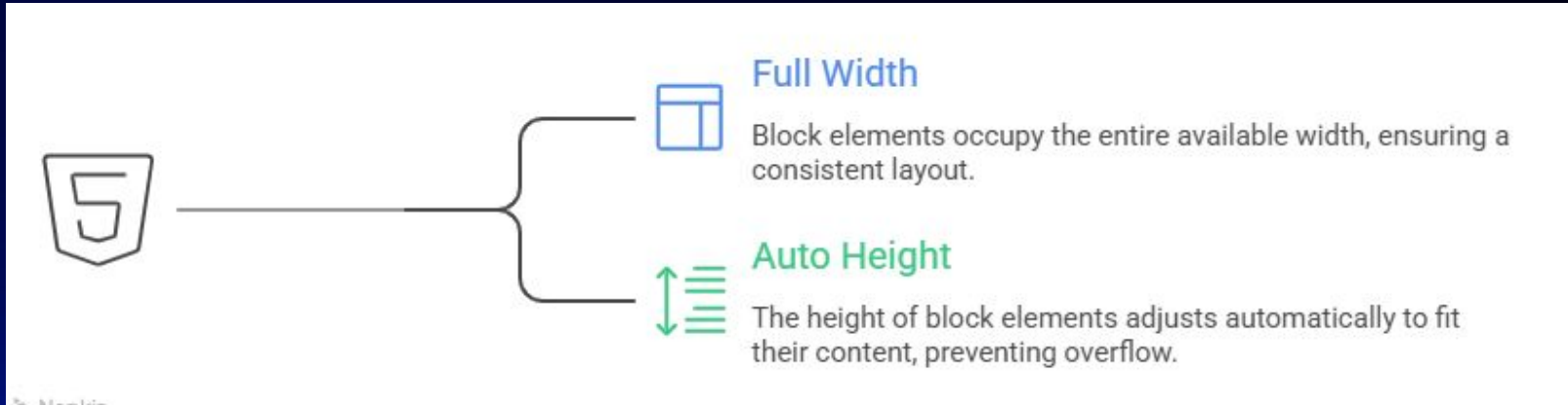
B

Browser DevTools



Width and Height

Default Behavior:



Key Properties:

Set Height

Controls the vertical size of the element



Set Width

Controls the horizontal size of the element



Applying width to sections

```
<section class="container">
  <h2>Welcome</h2>
  <p>This is a section with controlled
    width.</p>
</section>
```

```
.container {
  width: 60%;
  margin: 0 auto; /* centers the section */
  border: 1px solid black;
  padding: 20px;
}
```

Welcome

This is a section with controlled width.

Setting width for images

```

```

```
.responsive-img {  
  width: 100%;  
  max-width: 400px;  
}
```

- Avoid fixed **height** when possible
- Fixed height can cause content overflow or layout issues

Hands-On Demo | Width and Height

Q. What is the default width of a block-level element?

A

Auto (takes full available width)

B

Depends on height

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A

Auto (takes full available width)

B

Depends on height

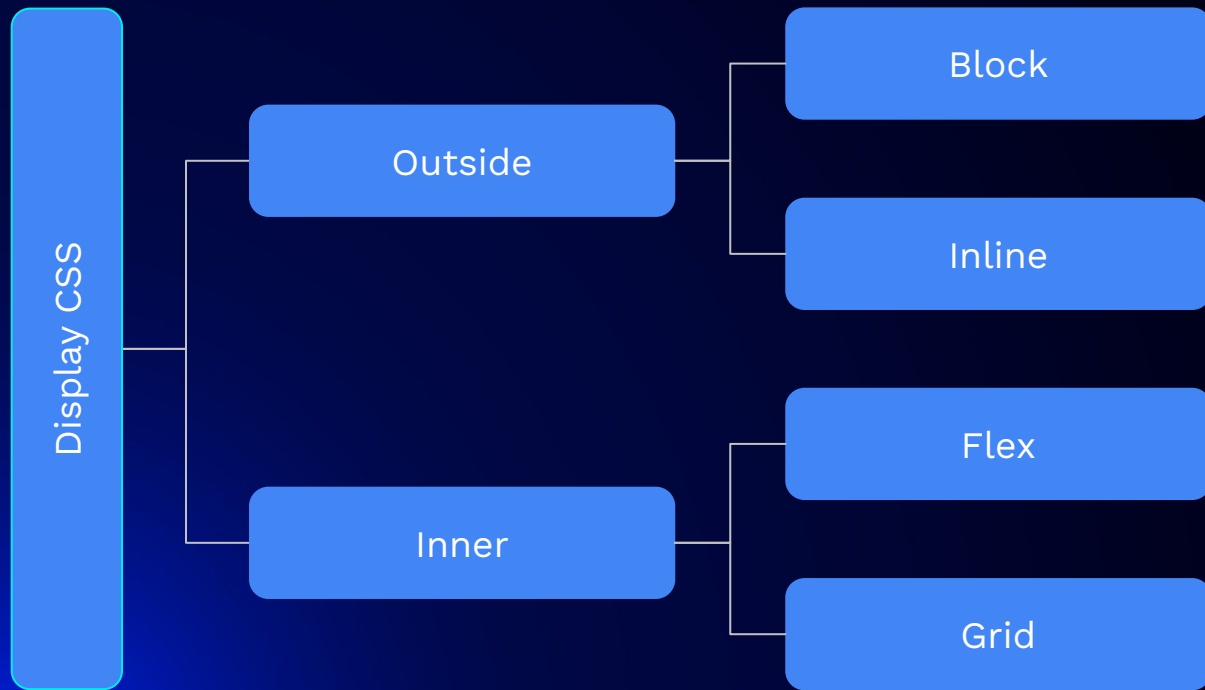


Display Property

Display

- It is a CSS property sets whether an element is treated as a block or inline box and layout used for its children, such as flow layout, grid or flex.
- We can alter the default layout, to some other layout options available using the display property as follows
- There are two kinds of display types
 - **Outside:** The outer display type of an element, can be inline, block, inline-block, or So, which determines its role in the flow layout
 - **Inside:** The inner display type defines how the children's elements will be arranged within the container element.

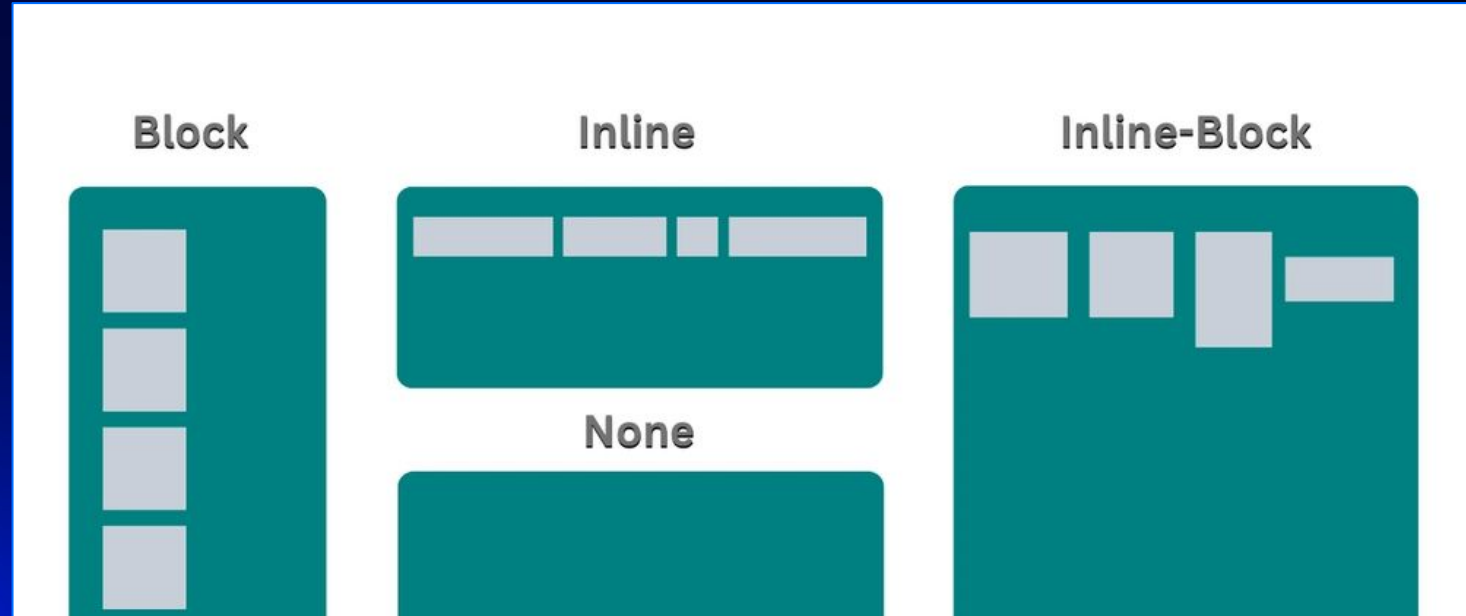
Display



CSS display property

Property	Description	Takes Up Full Width?	Starts on a New Line?	Examples
block	Elements are displayed as blocks.	Yes	Yes	<div>, <p>, <h1>
inline	Elements are displayed inline with the content around them.	No	No	, <a>,
inline-block	Elements are displayed inline but behave like block elements.	No	No	, <button>, <input>
none	Elements are not displayed at all.	No	No	(Elements set to display: none)

CSS display property



Hands-On Demo | Display Property

Pop Quiz

Q. Which of the following CSS display property is use to make an element behave like an inline element while retaining some features of a block-level element. ?

A

display:block

B

display:inline-block

Pop Quiz

Q. Which of the following CSS display property is use to make an element behave like an inline element while retaining some features of a block-level element. ?

A

display:block

B

display:inline-block



Visibility vs Display None

Display none

- `display: none` is a CSS property value that completely removes an element from the page layout,
- meaning it is not visible and does not take up any space.



Visibility

- It is a css property that shows or hides an element without changing the layout of a document.
- It can have the value of visible, hidden and collapse

```
<body>
  <div>
    
    </div>
  </body>
```

+

```
div {
  border: 1px solid
  red;
  width:
  fit-content;
}
img {
  visibility:-----;
}
```

* the collapse keyword has different effects for different elements mainly use in the table element in html

Visibility:
visible



Visibility: hidden



Difference between Display and Visibility

display: none;	visibility: hidden;
It is completely removed from the layout flow. It does not occupy any space on the page.	It becomes invisible, but it still occupies space in the layout.
Child elements inside an element with display: none are also hidden and not rendered.	Child elements inside an element with visibility: hidden are also hidden, but they still occupy space.

Hands-On Demo | Visibility vs Display None

Pop Quiz

Q. Which property will cause other elements to shift up when applied?

A

visibility: hidden

B

display: none

Pop Quiz

Q. Which property will cause other elements to shift up when applied?

A

visibility: hidden

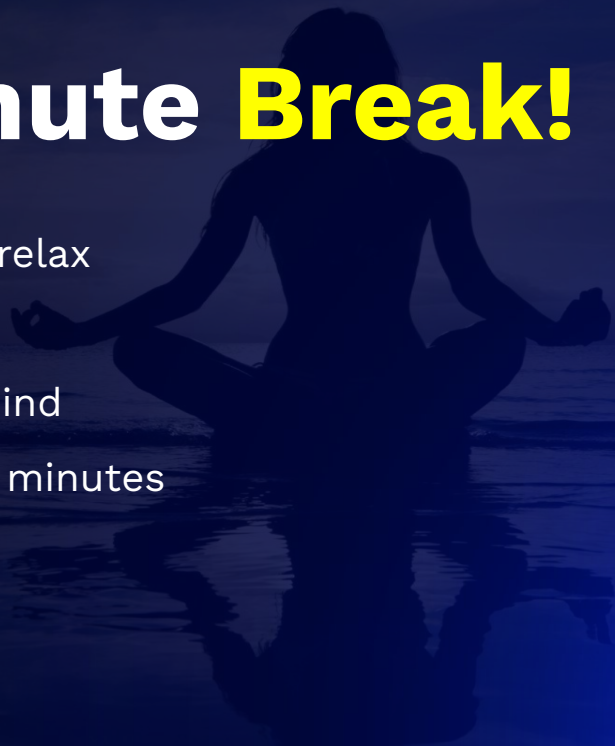
B

display: none



Take A 5-Minute **Break!**

- Stretch and relax
- Hydrate
- Clear your mind
- Be back in 5 minutes



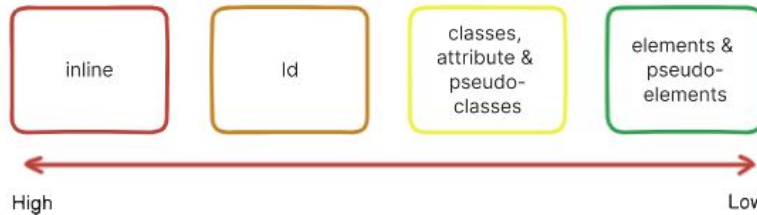


Basic Specificity (Simple Level)

Understanding CSS Specificity

The CSS specificity refers to the level of importance or weight assigned to a particular style rule.

CSS Specificity



How to calculate Specificity Order

CSS specificity is calculated based on the type of selector used and is represented by a four-digit value.

Note -

- Inline style gets 1000
- Type selector gets 1
- Class selector gets 10
- Id selector get 100
- Universal selector get 0

SELECTOR	SPECIFICITY VALUE	CALCULATION
p	1	1
p.test	11	1 + 10
p#demo	101	1 + 100
<p style="color:pink; >	1000	1000
#demo	100	100
.test	10	10
p.test1.test2	21	1 + 10 + 10
#navbar p#demo	201	100 + 1 + 100
*	0	0 (the universal selector is ignored)

Hands-On Demo | Basic Specificity (Simple Level)

Pop Quiz

Q. Which CSS selector has the highest specificity?

A

class selector

B

id selector

Pop Quiz

Q. Which CSS selector has the highest specificity?

A

class selector

B

id selector



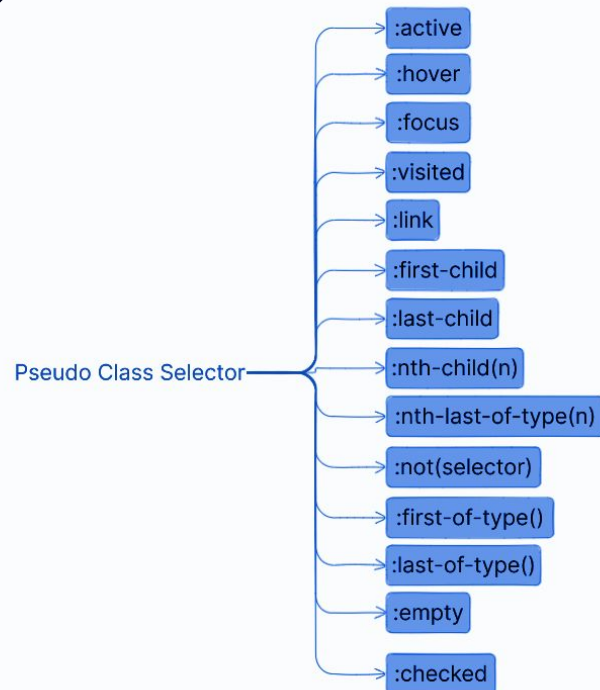
Pseudo-Classes (Interaction)

Pseudo class selector

Pseudo-classes are predefined keywords that select an element based on its state or target a specific child. they start with a single colon (:).

Syntax of Pseudo class selector -

```
tag:pseudo-class {  
  /* ...set CSS styling properties */  
}
```



style.css

```
a:active {  
    color: red;  
}  
/* This will change the anchor HTML tag to  
red on click of the link.*/
```

The Pseudo class selector “**:active**”, select an element when it is clicked or pressed down by the mouse pointer.

Browser output

Before click

Click the link to check how [active pseudo-class selector works](#)

After click

Click the link to check how [active pseudo-class selector works](#)

The Pseudo class selector “**:hover**”, select elements that are hovered by the mouse.

style.css

```
a:hover {  
    color: red;  
}  
/* On hover of the "PW  
Skills" the text color will  
change to red color */
```

Browser output

Before hovering

Hover the below link

PW Skills

After hovering

Hover the below link

PW Skills

The Pseudo class selector “**:focus**”, select elements that have received focus.

style.css

```
a:hover {  
    color: red;  
}  
/* On hover of the "PW  
Skills" the text color will  
change to red color */
```

Browser output

Before hovering

Hover the below link

PW Skills

After hovering

Hover the below link

PW Skills

Hands-On Demo | Pseudo-Classes (Interaction)

Pop Quiz

Q. Which pseudo-class is applied when an element is being clicked?

A

:active

B

:focus

Pop Quiz

Q. Which pseudo-class is applied when an element is being clicked?

A

`:active`

B

`:focus`

Class Project

Apply Styling to Project (Guided)

Summary

In this lecture, we have covered:

- Learned the Box Model for structure and spacing
- Understood width and height for sizing elements
- Explored the display property for layout behavior
- Compared visibility and display: none
- Introduced basic CSS specificity
- Used pseudo-classes for interaction

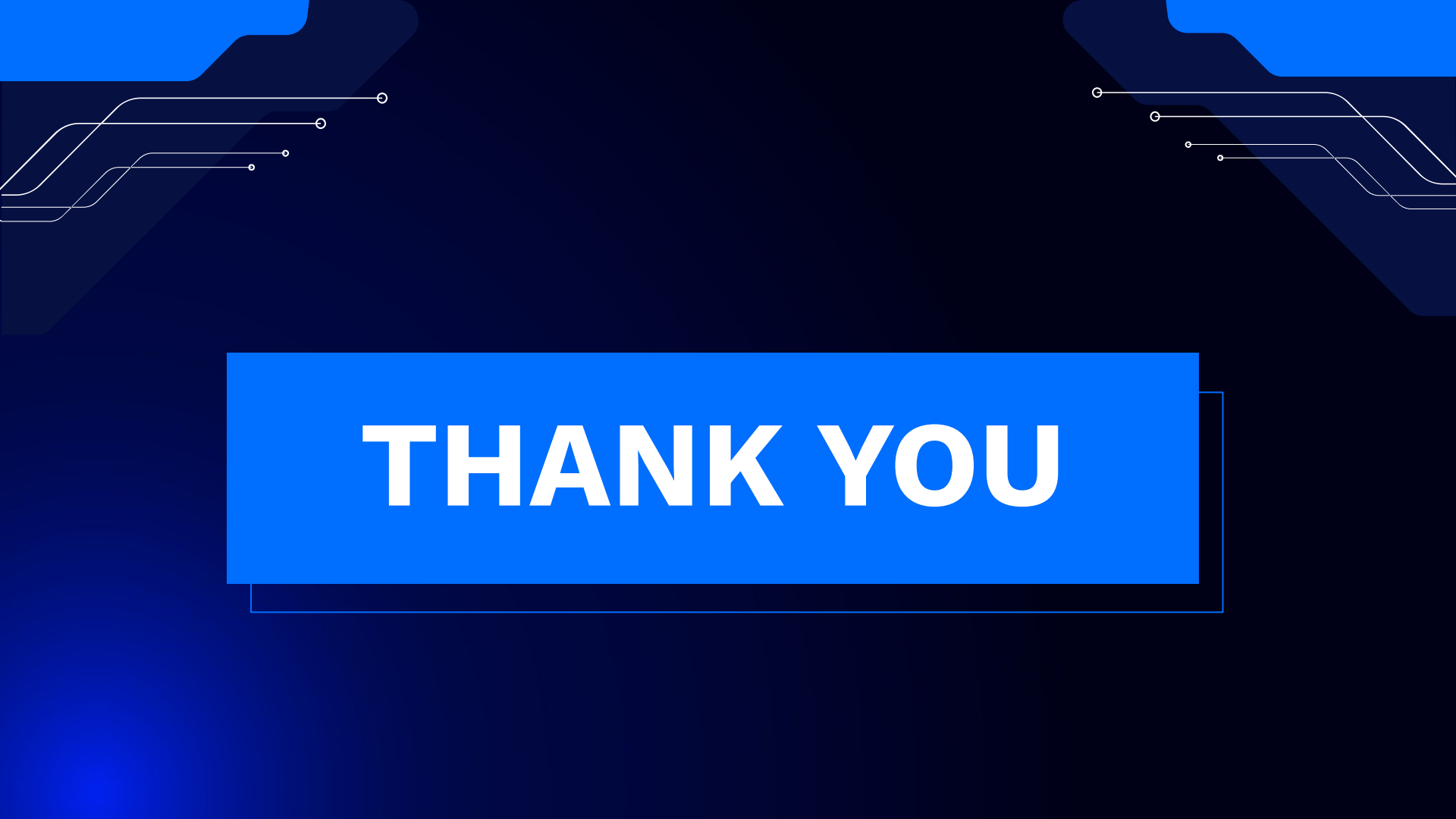




DOUBT RESOLUTION

Important

- Complete the post-class assessment
- Complete assignments (if any)
- Practice the concepts and techniques taught in this session
- Review your lecture notes
- Note down questions and queries regarding this session and consult the teaching assistants.



THANK YOU